❖Need Statement

In many temples, especially small or crowded ones, there is a shortage of priests to perform repetitive ritual tasks like ringing bells, lighting lamps, chanting pre-recorded mantras, and guiding devotees. This leads to delays, inconsistency, and difficulty in maintaining continuous rituals.

Therefore, a simple automated assistant (Poojari Bot) is needed to support routine temple activities safely and efficiently.

Objective

- Design a low-cost automated system that can:
- Ring temple bell at scheduled times
- Play devotional chants/mantras
- Light an electric diya (safe lamp)
- Provide basic guidance to devotees

⍰ Proposed Solution

A microcontroller-based robotic assistant installed near the temple altar that performs preset ritual support functions automatically or via button control.

Main Components

- Microcontroller (Arduino Uno or similar)
- Servo Motor → to ring bell
- Speaker module → play recorded mantra
- LED diya / electric lamp → safe lighting
- Push buttons / keypad → user control
- Power supply
- Frame/stand

Working Principle

- User presses a button or timer triggers action.
- Microcontroller sends signal to components.
- Servo rotates → rings bell.
- Speaker plays devotional audio.
- LED diya glows.
- System resets for next cycle.

Engineering Concepts Used

- Embedded systems
- Automation
- Human-machine interaction
- Basic robotics
- Safety design
- Product design thinking

Applications

- Small temples
- Home prayer rooms
- Festival automation
- Elderly assistance in daily rituals

Advantages

- Reduces manual repetitive work
- Low cost and easy to build
- Safe (no real fire used)

- Works on schedule
- Can run continuously

Limitations

- Cannot replace human priest
- Limited functions
- Needs electricity

Future Improvements

- Voice control system
- Mobile app integration
- Automatic incense dispenser
- Face detection for greetings
- Multilingual devotional audio
- Timer based sheduling

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